



Cassava is a staple food in many tropical regions of the world.

History & Folklore Bitter cassava has large quantities of highly toxic glycosides, and must be carefully soaked and cooked before it is safe to eat. (Sweet cassava is safe to eat without such processing.) *Tapioca* is a native Brazilian name for the processed root, which is used in commercial food preparation as a thickening agent. The Witoto of the Colombian Amazon poison fish with the water used to wash bitter cassava. The Makuna use the wash water to treat scabies.

Medicinal Actions & Uses Cassava root is easily digestible and makes a suitable, if low-protein, food for convalescence. The bitter variety may be used to treat scabies, diarrhea, and dysentery. Cassava flour may be used to help dry weeping skin. In China, a poultice is made of cassava, wheat flour, and ginger (*Zingiber officinale*, p. 155) to draw out pus when infection is present.

Caution Raw bitter cassava is toxic and has caused many deaths. The root must be carefully soaked and cooked before eating.

Maranta arundinacea (Marantaceae)

Arrowroot

Description Perennial growing to 6½ ft (2 m). Has a creeping rhizome, many long-stemmed oval leaves, and flowering stems with clusters of creamy-white flowers.

Habitat & Cultivation Native to northern South America and the Caribbean islands, arrowroot is cultivated mostly on the island of St. Vincent. The rhizome is unearthed 10 to 11 months after planting.

Part Used Rhizome.

Constituents Arrowroot contains 25–27% neutral starch.

History & Folklore In Central America, the Maya made the root into a poultice for smallpox sores, and an infusion for urinary infections. Arrowroot was a staple food of the Arawak people of the Caribbean. The plant reputedly gets its name from its use to treat poisoned arrow wounds—presumably as a drawing poultice.

Medicinal Actions & Uses Arrowroot is used in herbal medicine in much the same manner as slippery elm (*Ulmus rubra*, p. 145), as a soothing demulcent and a nutrient of benefit in convalescence and for those with weak digestions. It helps to relieve acidity, indigestion, and colic, and is mildly laxative. It may be applied as an ointment or poultice mixed with antiseptic herbs such as myrrh (*Commiphora molmol*, p. 85).

Self-help Use Acidity & indigestion, p. 307.

Marrubium vulgare (Lamiaceae) White Horehound

Description Square-stemmed perennial growing to about 20 in (50 cm). Has toothed, downy leaves and double-lipped white flowers.

Habitat & Cultivation Native to Europe, white horehound is naturalized in North and South America. It flourishes in dry, bare, or open areas, and is gathered in spring.

Parts Used Leaves.

Constituents White horehound contains the diterpenes marrubiin (0.3–1.0%) and marrubenol, tannins, and 0.06% volatile oil. Marrubiin is thought to be chiefly responsible for the herb's expectorant and bitter activity. It also acts on the heart to correct an irregular heartbeat.

History & Folklore White horehound has been a remedy for chest problems since ancient times, perhaps most frequently taken as a syrup made with honey or sugar. The Greek physician Dioscorides (40–90 CE) recommended a decoction of white horehound as a treatment for tuberculosis, asthma, and coughs.

Medicinal Actions & Uses White horehound is helpful for wheeziness, bronchitis, bronchiectasis (a damaged air passage within the lung), bronchial asthma, non-productive coughs, and whooping cough. The herb apparently causes the secretion of a more fluid mucus, which is readily cleared by coughing. As a bitter tonic, white horehound increases the appetite and supports the function of the stomach. It is widely used in Mexico to help control type 2 diabetes. The herb may also act to normalize heart rhythm, improving its regularity.



Dried herb

Marsdenia condurango syn. *Gonolobus condurango* (Asclepiadaceae) Condurango

Description Climbing vine growing to 33 ft (10 m). Has heart-shaped leaves and funnel-shaped, white-green flowers.

Habitat & Cultivation Condurango is native to deciduous forests of the Andes in Peru and Ecuador. It generally grows at altitudes between 3,300–6,600 ft (1,000–2,000 m). The bark is collected year round.

Parts Used Bark, latex.

Constituents Condurango bark contains glycosides (based on condurangogenins), a volatile oil, and phytosterols.

History & Folklore Early in the last century, condurango was erroneously yet widely believed to be a remedy for cancer.

Medicinal Actions & Uses The bark's main effect is to stimulate stomach secretions. It is often used in South American folk medicine as a potent bitter and digestive tonic. Condurango is a specific treatment for nervous indigestion and anorexia nervosa, since its bitterness slowly increases the appetite as well as the stomach's ability to process more food. The herb is also thought to stimulate the liver and pancreas, and may be taken for liver disorders. Condurango also encourages menstruation. The caustic white latex has been applied topically to remove warts.

Research The condurangogenins in condurango may act to counter tumors. The whole plant, however, does not seem to impede cancer development.

Caution The latex is poisonous and should not be taken internally.

Medicago sativa (Fabaceae) Alfalfa, Lucerne

Description Perennial herb growing to 32 in (80 cm). Has 3-lobed leaves, flowers that range in color from yellow to violet-blue, and spiraling seed pods.

Habitat & Cultivation Native to Europe, Asia, and North Africa, alfalfa is found in meadows and in both open and cultivated areas. Grown as a fodder crop in temperate climates, it is harvested in summer.

Parts Used Aerial parts, seeds, sprouting seeds.

Constituents Alfalfa contains isoflavones, coumarins, alkaloids, vitamins, and porphyrins. The isoflavones and coumarins are estrogenic.